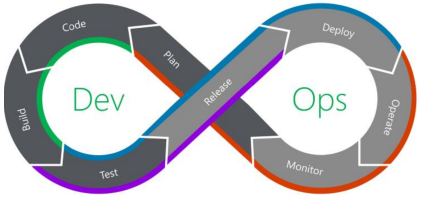
CldOps Cheatsheet

Yanick Egli & Doriano Di Pierro & Fabio Zahner

1 General Definitions

NIST Cloud: On-demand self-service, Broad network access, Resource pooling (multi-tenant), Rapid elasticity, Measured service (pay-as-you-go).
Cloud Operations: manage/optimize cloud services & infra.
GitOps: Git = single source of truth; declarative, versioned CI/CD.
DevOps: dev+ops; automation, CI/CD, monitoring, agile.
CI: developers integrate into the shared mainline frequently (small commits, short-lived branches); every push auto-triggers build + tests (unit/integration), catching conflicts/breakage early and keeping the trunk green & releasable. Avoids big-bang "integration hell"; fast feedback, fail fast.
CD: once CI passes, the change is auto-packaged & promoted through environments (dev→staging→prod). Continuous Delivery = automated up to a manual gate before prod (one-click release; artifact always deployable); Continuous Deployment = no gate, every green change auto-released to prod.
Platform Engineering: internal self-service dev platforms (IDP); faster shipping.
How they relate: DevOps (culture & practice) → CI/CD (automates it via pipelines) → Platform Engineering (productizes it as a self-service IDP to scale across teams).
DevOps 6 pillars: Cultural mindset, Automation-first, Continuous Processes, Monitoring/Feedback, Rapid+Reliable Releases, Security.

1.1 DevOps Cycle



Arcs → stages: CI: Code+Build+Test, **CDeployment:** Release+Deploy (+Test), **CDelivery:** Test+Release (manual prod), **CFeedback:** Operate+Monitor+Plan.
Stages (activities | tools):
Plan (requirements, roadmap, arch sketch, risk/rollback | Jira, Asana, GitHub Issues, Trello),
Code (write/refactor, local build/test, peer review | GitLab, GitHub, Bitbucket),
Build (compile, container images, publish artifacts, versioned packages | GitLab CI, GH Actions, Jenkins, Maven/Gradle, Docker/Podman, Artifactory, Quay),
Test (unit/integration/E2E, perf/load, sec scans, coverage, quality gates | JUnit/pytest, Selenium/Cypress/-Playwright, JMeter/k6, Snyk/Trivy, SonarQube),
Release (promote artifact, immutable version, release notes, signing | Cosign, GnuPG, GitLab/GH Packages),
Deploy (pull artifact, apply manifest, strategy, post-deploy validation, rollback | ArgoCD, Flux, Helm, Kustomize, Terraform/OpenTofu, Ansible, Flagger),
Operate (troubleshoot, scale, patch, secret rotation, drift detect, backups, compliance | PagerDuty, Vault, Velero, HPA/VPA, Falco),
Monitor (SLI/SLO, dashboards, alerts, synthetic checks, correlate logs/metrics/traces | Prometheus, Grafana, Loki, Jaeger, ELK, Alertmanager).

CD Deployment Strategies:
- **Rolling:** gradual update, minimal downtime
- **Blue-Green:** two envs, fast switch
- **Canary:** small user group first
- **Feature Flag:** deploy, toggle later
- **Dark Launching:** real traffic, hidden from users

1.2 Platform Engineering

Core Responsibilities: enable self-service, standardize runtimes/tooling, abstract infra, enforce compliance & security, provide observability.
8 Building Blocks (top→bottom): Self-Service Portal (IDP), Cost & Capacity, Observability, Policy & Security, Package & Artifact Mgmt, CI/CD Pipelines & Deployment, Infrastructure as Code, Compute & Orchestration.
Top Container Challenges (CNCF 2025): Cultural changes (47%), Lack of training (36%), Security (36%), CI/CD (35%), Monitoring (35%), Complexity (34%), Scaling, Networking, Logging, Service Mesh.

2 GitLab Pipelines

```
default: { image: golang:1.24-alpine } #
      fallback image for all jobs
stages: [build, test, release, deploy, post-
      deploy]

build: # DevOps cycle -> Build
  stage: build
  script:
    - go mod download
    - CGO_ENABLED=0 go build -o bin/app ./...
  artifacts: { paths: [bin/app] } # persist for
       later stages

lint-test: # DevOps cycle -> Test
  stage: test
  image: golangci/golangci-lint:v1.64.5-alpine
  script: [golangci-lint run -v]

build-docker: # DevOps cycle -> Release (
      publish image)
  stage: release
  image: docker:28
  services: [{ name: docker:28-dind, alias:
       docker }] # dind
  variables: { IMG: $CI_REGISTRY_IMAGE:
       $CI_COMMIT_SHORT_SHA }
  before_script:
    - docker login -u $CI_REGISTRY_USER -p
       $CI_REGISTRY_PASSWORD $CI_REGISTRY
  script:
    - docker build -t $IMG -f Dockerfile.prod .
    - docker push $IMG

deploy-prod: # DevOps cycle -> Deploy
  stage: deploy
  image: alpine/kubectl:1.35
  script:
    - yq -i ".spec.template.spec.containers[0].
       image=\"$IMG\"" k8s/deployment.yaml
    - kubectl apply -f k8s/deployment.yaml
  environment: { name: production, url: https://
       demo.example.com }
  rules: [{ if: '$CI_COMMIT_BRANCH == "main"' }]
       # prod only on main

post-deploy: # DevOps cycle -> Operate/Monitor
  stage: post-deploy
  script: [echo "smoke test + health check"]
```

2.1 Core Concepts & DSL Mapping

Pipeline = automated steps → faster, safer, reliable. Traceable logs, parallel jobs, consistent envs kill "works on my machine.
5-stage flow: commit/MR/tag → *build* (compile, image, package) → *test* (unit/integration/security/perf) → *release* (publish to registry) → *deploy* (target env) → *post-deploy* (smoke, health, rollback).
Map to DevOps cycle: the pipeline automates the loop's middle – `build`→Build, `test`→Test, `release`→Release, `deploy`→Deploy; *triggered by* Code (commit/MR/-tag), *feeds* Operate/Monitor. **CI** = build+test; **CD** = release+deploy.

2.2 Jobs, Stages & Rules

`.gitlab-ci.yml` YAML DSL. **Job** = smallest unit, belongs to one **stage**, runs `script` on a runner. Same stage → *parallel*; stages → *sequential*.
rules: *when* a job runs (modern replacement for `only`/`except`). **needs:** builds DAG, start before stage finishes. **dependencies:** which jobs' artifacts to pull (no ordering). **workflow:rules:** gate the whole pipeline.

```
workflow:
  rules:
    - if: $CI_PIPELINE_SOURCE == "
         merge_request_event"
    - if: $CI_COMMIT_BRANCH &&
         $CI_OPEN_MERGE_REQUESTS
      when: never
    - if: $CI_COMMIT_BRANCH
deploy:
  stage: deploy
  needs: [build] # DAG, skip stage wait
  dependencies: [build] # pull build's artifacts
  rules:
    - if: '$CI_COMMIT_BRANCH == "main"'
    - when: never
```

2.3 Cache, Artifacts & Services

Cache: shared across jobs/pipelines, speed up repeat work (deps, compiled objects).
Artifacts: files persisted to GitLab for downstream stages or download; declare via `dependencies`/`needs`.
Services: sidecar containers (DB, dind, API) reachable via `alias`.

```
integration-test:
  services: [{name: postgres:17, alias: db}]
  variables: {POSTGRES_DB: app, POSTGRES_USER: u
       , POSTGRES_PASSWORD: p}
  script: [psql -h db -U $POSTGRES_USER -d
       $POSTGRES_DB -c "SELECT 1"]
  cache: {key: deps, paths: [.cache/]}
  artifacts: {paths: [bin/], expire_in: 1h}
```

2.4 Environments

Where code is deployed. **Tiers:** Local (workstation) → Dev (sandbox) → Integration (CI target) → Test/QC (interface tests) → Staging (prod mirror) → Production (live). Linkable to K8s cluster (set up via UI). `environment.on_stop` unstages; `action: stop` stops it.

2.5 Push vs. Pull Deployment

Push (Jenkins/GitLab calls cluster): **pro:** easy, flexible targets (VMs/containers, not just K8s); **con:** firewall open, admin access out, pipeline per env.
Pull (agent in cluster, e.g. Argo/Flux): **pro:** secure (no inbound), auto-reconcile, scales identically; **con:** agent per cluster, polling latency.

2.6 GitLab Runner

Worker executing jobs defined in .gitlab-ci.yml. **Scope:** Instance (shared, GitLab-managed) / Group / Project (dedicated). **Executors:** `shell`, `ssh`, `docker`, `docker+machine`, `kubernetes`, `virtualbox`, `parallels`. Register: `gitlab-runner register` with URL + token + tags + executor. Jobs pick runner by `tags:`.

2.7 More Keywords & Predefined Vars

`default:` fallback `image/before_script`. `before_script/after_script` wrap every `script`. `rules:changes:` run only if files changed.
Predefined vars:
`$CI_REGISTRY[_IMAGE]`, `$CI_REGISTRY_USER/_PASSWORD`, `$CI_COMMIT_BRANCH`, `$CI_COMMIT_SHORT_SHA`, `$CI_PROJECT_DIR`, `$CI_JOB_TOKEN` (auth API/registry), `$CI_MERGE_REQUEST_IID`. Pin images by digest (`img@sha256:..`), not `:latest` → flaky.

3 Terraform

IaC: manage & provision infrastructure (compute, storage, network) through code instead of manual processes (Red Hat).
Motivation: replace slow, drift-prone manual console clicking with code; Automation pays off but carries its own upkeep cost.
Benefits: idempotent (declarative, re-apply = same result), drift detection (state vs. reality), version-controlled (history/review/rollback), consistency (change a source → all references update), reusable & standardized (modules: write once, deploy many, enforce best practices), team collaboration (shared, locked remote state), flexible (variables, no hard-coded secrets).
Declarative vs. Imperative: Terraform is declarative – you describe the desired end state and TF figures out the steps to get there. Imperative (shell scripts, manual CLI calls) instead describes how to reach a result step by step.
Toolchain (4 components): language (HCL – declare infra & vars), state (`.tfstate`, maps config to real resources), provisioner = Terraform-core plugins: Providers (access a cloud's API: AWS/Azure/GCP/K8s) + Provisioners (`remote-exec`/`local-exec` scripts on a resource), backend (where state lives: local or remote S3+lock; `init` sets it up + downloads plugins).
TF doesn't speak directly with an SDK, but rather Terraform -> Provider -> Client SDK. Diffrent providers enable diffrent platforms (AWS, Azure, Kubernetes, ...). **Resources** (`resource`) are the "whatTTF creates and manages (e.g. an EC2 instance). **Data Sources** (`data`) are read-only lookups of existing infrastructure used as input (e.g. an existing AMI or VPC). A sample in HCL (Hashicorp Configuration Language):

```
terraform {
  required_version = ">= 1.0.0"
  required_providers { aws = { source = "
       hashicorp/aws", version = "~> 5.0" } }
}
provider "aws" { region = "us-east-1" }
variable "instance_type" { default = "t2.micro"
      }
resource "aws_instance" "web" {
  ami = data.aws_ami.ubuntu.id
  instance_type = var.instance_type
}
output "public_ip" { value = aws_instance.web.
      public_ip }
```

To deploy infrastructure, write HCL in files like `main.tf`, then run `terraform init`, `terraform plan` to show changes that would be made, `terraform apply` to actually apply the changes. Use `terraform destroy` to delete all made changes.

3.1 OpenTofu

Fork of Terraform v1.5.7 under **Linux Foundation**, MPL-2.0 (TF switched to BUSL). Drop-in replacement, adds **state encryption**. Avoids single-vendor lock-in.

3.2 HCL Features

count (numeric loop, `count.index`) and **for_each** (map/set loop, `each.key`/`value`) for multiple instances. **Data source** example:

```
data "aws_ami" "ubuntu" {
```

```
  most_recent = true
  filter { name = "name" values = ["ubuntu/
        images/hvm-ssd/ubuntu-*"] }
  filter { name = "virtualization-type" values =
        ["hvm"] }
  owners = ["099720109477"] # Canonical
}
```

3.3 Variables

Precedence: CLI `-var/-var-file` > env `TF_VAR_<name>` > `*.tfvars` > default. No default & not supplied -> CLI prompts operator. `TF_VAR_` prefix ideal for secrets (`TF_VAR_access_key=...`). `depends_on` forces explicit ordering; implicit deps come from attribute references.

3.4 State

`terraform.tfstate` (JSON) maps HCL to real resource IDs (e.g. ARN). **3-way diff** on apply: **NEW** (HCL) vs **PREVIOUS** (.tfstate) vs **EXISTING** (cloud refresh) -> decides actions, detects drift. **Remote backend** (S3) shares state in teams; **state locking** via `use_lockfile` (S3-native, replaces DynamoDB):

```
terraform {
  backend "s3" {
    bucket = "my-tfstate"
    key = "prod/terraform.tfstate"
    region = "us-east-1"
    use_lockfile = true
  }
}
```

3.5 File & Folder Layout

```
├─ modules/
│      shared modules, reusable across envs
├─ dev/
│  ├─ terraform.tf
│  │  required_version + providers
│  ├─ backend.tf
│  │  remote state (kept out of VCS)
│  ├─ providers.tf
│  │  provider auth/config
│  ├─ main.tf
│  │  resources & data sources
│  ├─ variables.tf
│  │  inputs (alphabetical)
│  ├─ outputs.tf
│  │  returns (alphabetical)
│  ├─ locals.tf
│  │  derived values
│  └─ override.tf
│     loaded last (use sparingly)
└─ prod/
       same layout, its own state
```

Per-env folders → separate state, so a corruption hits only one env.

3.6 Modules

Reusable folder of `.tf` files (root config is itself a module):

```
module "vpc" {
  source = "./modules/vpc" # or registry/git URL
  cidr = "10.0.0.0/16" # input variable
} # reference an output: module.vpc.vpc_id
```

3.7 Patterns & Recovery

`import` adopts existing infra; `terraform apply -replace=<addr>` (or legacy `taint`) forces a destroy+recreate of a single resource. Region-aware AMI via map:

```
import { to = aws_instance.web id = "i-0abc123"
       } # then: apply

variable "AMIS" { type = map(string) default =
      { us-east-1 = "ami-05b1..." } }
ami = var.AMIS[var.region]
```

4 Ansible

Configures provisioned machines (install/configure software, deploy code); *after* provisioning. Idempotent (no-op if already in desired state), agentless, push-based, no statefile/locking. Needs Python on host & targets. Other CM tools: **Puppet**, **Chef**, **CFEngine**, **Juju**.
TF vs Ansible: TF = provisioning, declarative, state, idempotent. Ansible = CM, *hybrid* (imperative tasks, declarative modules), no lifecycle, idempotency *optional*. Both: templating, agentless, OSS. Often combined.
Execution: over SSH, copies generated Python module to `~/.ansible/tmp` on target, runs with remote Python, returns JSON on stdout, deletes. Nothing stays installed. **Exceptions:** `raw` module runs cmd via SSH (no Python needed, e.g. bootstrap); `connection: local` runs on controller (network devices); Windows via **WinRM**.

4.1 Initial Setup & Inventory

`ssh-keygen -t ed25519 -C *@y"` on controller, `ssh-copy-id user@server`, test with `ansible all -m ping`.
Inventory (INI or YAML) lists managed hosts grouped, referenced by play's `hosts:`. `ansible-inventory -i inv --list` verifies it.

```
[web]
web1.example.com
web2.example.com ansible_host=10.0.0.5
[web:vars]
ansible_user=deploy
```

4.2 Playbooks, Plays & Tasks

Hierarchy: a playbook is one YAML file holding 1..n plays; a play maps a group of hosts to an ordered list of tasks; each task is a single module call (e.g. apt, copy), the atomic unit of work.
Run: `ansible-playbook -i inv playbook.yml` (`-i` = inventory file). Handy flags:

- `–check`, dry run: report what *would* change, change nothing
- `–list-tasks`, list the tasks without running them (preview)
- `–tags <t>` / `–skip-tags <t>`, run only / skip tasks tagged `<t>`
- `–ask-vault-pass`, prompt for the Vault password (decrypt vaulted vars)

```
- name: Example Playbook
  hosts: web
  become: true # sudo
  gather_facts: false # skip setup (faster)
  force_handlers: true # run handlers even if a
        task fails
  vars:
    packages: [nginx, curl]
    secret: "{{ vault_password }}"
  roles:
    - myrole
  tasks:
    - name: Install packages # list-as-option >
          loop (faster, deps OK)
      ansible.builtin.apt: # FQCN: namespace.
            collection.module
        name: "{{ packages }}" # alternative for
            apt is yum
        state: present
      notify: restart nginx
    - name: Render config
      ansible.builtin.template: { src: app.conf.
            j2, dest: /etc/app.conf }
      when: enable_service
      tags: config
  handlers:
    - name: restart nginx
      ansible.builtin.service: { name: nginx,
            state: restarted }
```

Handlers: special tasks that run *only when notified* (by a task reporting changed) and *once*, at the **end of the play** – even if notified by many tasks. Ideal for "restart/reload after a config change". Triggered with `notify: <handler name>`; a failed play skips pending handlers unless `force_handlers: true`. **Loop anti-pattern:** a per-item `loop:` runs the module once per package – slower & can break on interdependencies; prefer the whole list in one `name:` (single transaction, as above):

```
- name: suboptimal # one apt/yum call per item
  ansible.builtin.apt: { name: "{{ item }}",
        state: present }
  loop: "{{ packages }}"
```

4.3 Modules, State, Ad-hoc & Errors

Modules: the units of work; most are *idempotent* – you declare the desired `state:` and the module changes reality only if needed, returning ok/changed/failed as JSON. Common: apt/yum (pkgs), copy/template (files), service/systemd, file (dirs/-links), user, git, uri.
State: Ansible is *stateless* – no state file (unlike TF). Idempotency comes from each module comparing the desired state (`present/absent/started/stopped/latest`, ...) to reality on every run, so re-running converges. `--check` = dry run, `--diff` = show what changes.
Ad-hoc (one module, no playbook): `ansible -i inv -m <module> -a '<args>' <target>`. The setup module gathers *facts* (what "Gathering Factsrruns before tasks). `command` vs `shell`: `shell` allows pipes/redirects/vars; *neither is idempotent*. `ansible-inventory -i inv --list` verifies the inventory; `--check` is a dry run. A playbook stops at the first failed task unless `ignore_errors: true`.

```
- name: validate
  ansible.builtin.command: httpd -t
  register: out
  changed_when: out.failed
  failed_when: false
- ansible.builtin.file: { path: /data, state:
      directory } # file|link|touch|absent
- ansible.builtin.lineinfile: { path: /etc/x.
      conf, line: "Include sites/*" }
```

4.4 Vault, Collections, Roles & Tags

Vault: `ansible-vault create|edit|view|encrypt foo.yml`; `encrypt_string 'secret' --name 'pw'` inlines a single var. Pass via `--vault-id @prompt|/path/file|name@script`. Ciphertext safe to commit.
Collections are bundles of modules/roles/plugins. FQCN `ns.coll.mod` (e.g. `cisco.ios.ios_banner`); independent versioning. Install: `ansible-galaxy collection install <n>` or via `requirements.yml` (`collections:` + `roles:`) then `install -r`. Sources: **Galaxy** (community), **Automation Hub** (RedHat, certified).
Roles: `ansible-galaxy init <n>` creates dirs `tasks/ handlers/ defaults/ vars/ files/ templates/ meta/ tests/`; use via `roles:` list.
Tags can be used to execute a subset of tasks instead of the whole playbook. Run only specific tags by appending `--tags <name>` at the end of the ansible-playbook command. There are also two special commands: Tag *always* runs every time, except when explicitly skipped: `--skip-tags=always`. Tag *never* does not run unless specified with `--tags=never`

4.5 Jinja2 Templating

Rendered on **controller**, only result sent to target. `{{ inventory_hostname }}` = current target. Filters: `ipaddr/ipv4/ipv6` (validate/version), `hash('sha1')`, `default(x)`, `min/max/shuffle`.

```
{% for host in groups['webservers'] %}
BalancerMember http://{{ host }}:{{ port |
    default(80) }}
{% endfor %}
```

5 Kubernetes (K8)

Problems solved: automate deployment, scaling & management of containerized apps across many nodes: doing it by hand (placement, restarts, updates, scaling) doesn't scale.

> **Object:** persistent record-of-intent. **Controller:** reconciles current → desired state (the *control loop*) → fault-tolerance (auto-replace failed pods), replica count (ReplicaSet), rolling updates + rollback (Deployment), autoscaling (HPA).

5.1 Architecture

Control plane: `kube-apiserver` (REST front), `etcd` (KV store), `kube-scheduler` (pod → node), `kube-controller-manager` (built-in controllers), `cloud-controller-manager` (cloud glue, optional). **Node:** `kubelet` (runs pods via container runtime), `kube-proxy` (Service networking/iptables).

5.2 Objects

Pod: smallest unit, 1+ containers share net/storage.
Sidecar: container running alongside main in same pod (logs, proxy).
Init container: runs to completion before main.
ReplicaSet: keeps N pod replicas (rarely used directly).
Deployment: manages ReplicaSets, rolling updates, rollback.
StatefulSet: ordered pods with stable identity + per-pod PVC (DBs, brokers).
DaemonSet: one pod per (matching) node: logs, CNI, monitoring.
Service: stable VIP/DNS + LB over pod set.
Ingress: L7 HTTP(S) entrypoint.
Volume: dir mounted into pod.
ConfigMap: non-secret KV config.
Secret: sensitive KV, base64-encoded, *not* encrypted by default.
HPA: autoscales replicas on metrics.
CRD (Custom Resource Definition): registers a new object *kind* to extend the K8s API; an **Operator** = custom controller that reconciles such custom resources (e.g. `ServiceMonitor`, Argo `Application`, `CiliumNetworkPolicy`).

```
apiVersion: apps/v1
kind: Deployment
metadata: { name: app }
spec:
  replicas: 3
  selector: { matchLabels: { app: web } }
  strategy:
    type: RollingUpdate
    rollingUpdate: { maxUnavailable: 1, maxSurge:
          1 }
  template:
    metadata: { labels: { app: web } }
    spec:
      initContainers:
      - { name: init, image: busybox, command: ['
            sh','-c','sleep 10'] }
```

```
    containers:
    - name: web
      image: nginx
      ports: [{ containerPort: 80 }]
    - { name: sidecar, image: busybox, command:
          ['sh','-c','while true; do sleep 30;
          done'] }
```

5.3 Services

Give stable virtual IP + DNS over label-selected pods, round-robin LB. **ClusterIP** (default) internal only; **NodePort** static port on every node `<NodeIP>:<port>`; **LoadBalancer** cloud LB w/ public IP; **ExternalName** CNAME alias to external DNS (no proxy). **Ingress** routes HTTP(S) by host/path (needs controller: Nginx, Traefik). **Gateway API** (`Gateway+HTTPRoute/TLSRoute`) – modern, multi-protocol, role-oriented Ingress replacement. *Port trap:* Service `port` vs `targetPort` (container) vs `nodePort`; Ingress backend refs Service port.

5.4 ConfigMaps & Secrets

Inject into Pods as a single env var (`configMapKeyRef/secretKeyRef`), all keys at once (`envFrom`), or mounted as files (volume).

```
env:
- name: APIKEY
  valueFrom: { configMapKeyRef: { name: cfg, key
      : api-key } }
envFrom:
- configMapRef: { name: cfg }
volumeMounts:
- { name: v, mountPath: /etc/cfg }
volumes:
- { name: v, configMap: { name: cfg } } # or:
    secret: { secretName: s }
```

5.5 Storage: PV, PVC, StorageClass

Container FS is ephemeral. **Volume:** dir in pod, lives w/ pod (`emptyDir`, `hostPath`, NFS, …). **PersistentVolume (PV):** cluster-scoped piece of external storage. **PersistentVolumeClaim (PVC):** namespaced request for storage (size, accessMode); binds to a matching PV. **StorageClass:** dynamic provisioner – creates a PV on-demand if no static PV matches (e.g. `gp3`, `standard-rwo`).

5.6 LimitRange & ResourceQuota

LimitRange: per-container min/max + default req/lim in NS; out-of-range Pod rejected, missing req/lim auto-injected. **ResourceQuota:** caps summed NS usage; over-quota Pod rejected.

5.7 Namespaces

Logical isolation in one cluster (RBAC, quotas). Namespaced (Pods, Services) isolated; cluster-wide (Nodes, PV) shared. Defaults: `default`, `kube-system`, `kube-public`, `kube-node-lease`.

5.8 Rolling Updates & Rollback

Deployment ensures availability during change. **Recreate:** kill all → recreate (downtime). **RollingUpdate** (default): tuned by `maxUnavailable` (max pods down) + `maxSurge` (max extra pods). **History & rollback:** Deployment records a revision history, `kubectl rollout history deploy/x` lists revisions, `rollout undo deploy/x –to-revision=N` reverts. **Reliable config over time:** imperative is fast but hard to track/maintain → keep object YAML in **Git** (version control); alternative rollback = restore the file from an earlier commit & `kubectl apply`.

5.9 Scheduling

`kube-scheduler` picks node via **Filtering** then **Scoring**.
Filtering (feasibility): PodFitsHostPorts, PodFitsResources, PodMatchNodeSelector, CheckNodeDiskPressure, CheckVolumeBinding.
Scoring (rank): SelectorSpreadPriority (spread same Service/STS/RS), LeastRequestedPriority (free resources), NodeAffinityPriority.

```
spec:
  nodeSelector: { disktype: ssd }
```

Taints on nodes repel pods;
Tolerations on pods allow (don't force) them onto tainted nodes – pair w/ `nodeSelector`/affinity to pin. Effects: *NoSchedule, PreferNoSchedule, NoExecute* (evict). No effect on DaemonSets.
CLI: `kubectl taint nodes <n> gpu=true:NoSchedule`, `kubectl label nodes <n> disktype=ssd` (trailing - removes).

```
tolerations:
- { key: "gpu", operator: "Equal", value: "true
    ", effect: "NoSchedule" }
```

cordon/drain: `kubectl cordon <n>` marks unschedulable (existing stay); `drain <n>` also evicts pods (maintenance); `uncordon <n>` reverts.

5.10 Probes & Resources

kubelet probes via `httpGet/tcpSocket/exec`.
Liveness: fail → restart.
Readiness: fail → remove from Service endpoints.
Startup: guards slow boots, disables others until OK. Repeated liveness fail → exp backoff → `CrashLoopBackOff`.
requests: reserved min (scheduling).
limits: ceiling: CPU throttled, memory over-limit → OOMKilled.

```
resources:
  requests: { cpu: "250m", memory: "64Mi" }
  limits: { cpu: "500m", memory: "128Mi" }
livenessProbe:
  httpGet: { path: /healthz, port: 80 }
  initialDelaySeconds: 5
  periodSeconds: 10
readinessProbe:
  httpGet: { path: /ready, port: 80 }
```

5.11 Autoscaling (HPA)

HorizontalPodAutoscaler: autoscales a workload's replica count on observed metrics, holds performance during traffic spikes, cuts cost at low usage.
Key config: minReplicas, maxReplicas, target metric (threshold that triggers scaling, e.g. 70% CPU).
Metrics: *resource* (CPU/memory util, needs Metrics Server), *custom* (app-specific via adapters), *external* (traffic-based, e.g. requests/s).
Best practice: pods must set resource requests/limits, HPA computes % against them (none → no scaling); test scaling under load before prod.

```
kubectl autoscale deploy php-apache --cpu-
    percent=50 --min=1 --max=10
```

5.12 Imperative vs Declarative

3 ways to manage objects:
Imperative commands: act on live objects straight from the CLI (`kubectl run/create/expose/scale/delete`); fast for one-off/learning, but not reproducible, no single source of truth, hard to review.
Imperative object config: file + explicit verb (`kubectl create|replace|delete -f f.yaml`); `replace` *overwrites* the object (drops changes not in the file).
Declarative config: describe desired state in YAML, K8s computes & applies the diff (`kubectl apply -f <dir/>`, even a whole folder); idempotent, repeatable, Git-friendly, `kubectl diff -f` previews. *Recommended* (course: more control than CLI-only).
Workflow: scaffold with imperative `--dry-run=client -o yaml`, then manage declaratively via `apply`.

5.13 Commands

Apply: `kubectl {create|apply|replace} -f manifest.yaml`
Gen YAML: `kubectl create deploy web --image=nginx --dry-run=client -o yaml > d.yaml`
Imperative: `kubectl run`, `scale deploy x --replicas=4`, `expose`
Update image: `kubectl set image deploy/x web=nginx:1.27`
Inspect: `kubectl get pods -A -o wide`, `describe node <n>`, `get nodes --show-labels`
Exec: `kubectl exec -it nginx-xxx -- sh` (drop -it for one-shot)
Rollout: `kubectl rollout {status|history} deploy x`, `rollout undo deploy x --to-revision=N`
Node ops: `kubectl cordon|uncordon|drain <n>`
Default ns: `kubectl config set-context --current --namespace=lab`

6 Helm

Package manager for K8 (apt/brew-like).
Chart: files describing related K8 resources.
Release: running instance.
Repo: stores versioned charts.
Architecture: client-side only (Helm 3, no Tiller); talks to K8 API directly; release state stored in-cluster as **Secrets** (in the release's namespace).
Pros: huge chart library, versioning/deps, lifecycle (install/upgrade/rollback), powerful Go-template logic.
Cons: templates hard to debug, "spaghetti YAML", opaque state, steep curve. Use for: distributing complex apps, consuming community charts, advanced logic. Avoid for: simple YAML tweaks.

6.1 Commands

```
helm repo add|list|rm NAME URL # repos
helm search repo <r> / <c> -l # all versions
helm install -f vals.yaml NAME CHART --version
      1.2.3
helm upgrade RELEASE CHART
helm rollback RELEASE REVISION
helm uninstall RELEASE_NAME # also drops
      history
helm list; helm history RELEASE_NAME
helm status RELEASE_NAME # unknown|deployed|
      uninstalled|
  # superseded|failed|uninstalling|pending-{
      install,upgrade,rollback}
helm get all|values|notes|manifest
      RELEASE_NAME # values=overrides
helm template NAME CHART -f v.yaml --set k=v #
      render local
helm install NAME CHART --dry-run --debug #
      simulate
helm lint CHART_PATH [--with-subcharts]
helm show chart|readme|values|all CHART
helm package CHART_PATH # build .tgz, then helm
      push <pkg> <repo>
helm dependency update CHART # pulls into
      charts/
```

6.2 Chart Structure

`helm create NAME` scaffolds:
Chart.yaml metadata
values.yaml default values
templates/ Go-template manifests
templates/_helpers.tpl named templates (_=not rendered)
templates/NOTES.txt rendered & shown post-install
charts/ subcharts; crds/ CRDs
.helmignore ignore globs (!=negate)

6.3 Chart.yaml fields:

apiVersion v2=Helm 3+, v1=legacy
name chart name;
version chart version, SemVer2 (e.g. `1.4.0`) → `name-version.tgz`
type `application` (default) | `library` (helpers only, no install)
appVersion app shipped (info-only)
kubeVersion allowed range (e.g. `>=1.12 <1.14`); install fails if mismatch
dependencies name/version/repository/condition/alias (`requirements.yaml` deprecated; lockfile on update)

6.4 Templating

Objects: .Release.{Name, Namespace, Revision, IsInstall, IsUpgrade}, .Chart.*, .Values, .Files.Get, .Capabilities, .Template.
Values precedence (low→high, last wins): subchart values.yaml → parent values.yaml → `-f my.yml` → `--set k=v`; unset via `--set x=null`.
Functions (Sprig, ~60+): upper, quote, default, trunc, toYaml, indent/nindent. | pipes result as last arg. include = pipeable template. required fails if empty. tpl renders a string as a template.
Flow: `{{if}}…{{end}}` (false on `0//nil/empty`); `{{range $k,$v := .list}}`; `{{with .sub}}` rescopes `.`
Whitespace: `{{-` trims left, `-}}` trims right; `nindent N` = newline+indent. YAML indent is critical.

```
{{- define "app.name" -}}{{ default .Chart.Name
      .Values.name | trunc 63 }}{{- end }}
name: {{ include "app.name" . | quote }}
{{- required "set .Values.img!" .Values.img }}
resources: {{- toYaml .Values.resources |
      nindent 4 }}
```

7 Kustomize

Template-free overlay tool, built into kubectl (`-k`). `kustomization.yaml` lists **resources** + declarative **transformers** (cross-cutting fields) + **patches** (partial spec edits) + **generators**.
Pros: pure YAML, native to kubectl, easy diff/review, easy to learn.
Cons: less powerful than Helm, no deps mgmt, hard to package/distribute, limited conditionals.
Use for: env-specific configs (dev/prod), avoiding templating. Often combined with Helm.

7.1 Commands

```
kustomize create --resources deploy.yaml,svc.
      yaml # scaffold
kustomize edit add resource|set image|add label
      |remove resource
kustomize build <dir> # render to stdout
kustomize build <dir> | kubectl apply -f -
kubectl apply -k <dir> # native
kubectl kustomize <dir> # render only (no apply
      )
kubectl delete -k <dir>
kustomize cfg fmt <dir> # format yaml
kustomize localize <dir> # bundle remote URLs/
      git locally
kustomize edit fix # auto-migrate deprecated
      syntax
```

7.2 kustomize.yaml

Deprecated → new:
`commonLabels` → `labels`
`patchesStrategicMerge/patchesJson6902` → `patches`
`bases:` → inside `resources:`

```
apiVersion: kustomize.config.k8s.io/v1beta1
kind: Kustomization
resources: [../../base, storage.yaml] # files,
      dirs, URLs
namespace: prod # transformers
namePrefix: dev-
```

```
nameSuffix: -v1
commonAnnotations: {version: 1.0.0}
labels: # replaces commonLabels
- includeSelectors: true
  pairs: {app.kubernetes.io/name: hello}
images: # change tag/name w/o editing
      Deployment
- {name: nginx, newName: bitnami/nginx, newTag:
        latest}
patches: # strategic or JSON6902
- target: {kind: Service, name: nginx-app}
  patch: |-
    - {op: replace, path: /metadata/name, value:
          nginx-server}
configMapGenerator: # hash suffix appended ->
      change forces rolling restart
- name: app-config
  literals: [LOG_LEVEL=debug]
  files: [config.env] # or files:
secretGenerator: # base64-encodes; same hash
      mechanism
- {name: ex-secret, literals: [user=admin, pass
      =secret]}
components: [../../components/mysql]
```

ConfigMap/Secret modes: (1) *Traditional*: declare as resource → change does **not** restart pods (silent drift). (2) *Generator*: `configMapGenerator`/`secretGenerator` appends content hash to name → references rewritten → pods rolling-restart on change.

7.3 Bases, Overlays, Components

Base: common (`base/` with `kustomization.yaml`). **Overlay:** env-specific dir (`overlays/dev`, `overlays/prod`) that lists base in `resources:` + adds patches/transformers → produces a **variant**. **Component:** `kind: Component` (`v1alpha1`); reusable mix-in chunk (resources+patches+generators), included via `components:` in overlays. `resources` = full manifests/bases; `bases:` key itself is deprecated.

```
# components/mysql/kustomization.yaml
apiVersion: kustomize.config.k8s.io/v1alpha1
kind: Component
resources: [deployment.yaml, service.yaml]
```

7.4 Best Practices & Pitfalls

Keep common values (namespace, labels) in base. Organize by kind (`services.yaml`, `hpa.yaml`). Standard dirs: `base/`, `overlays/`, `components/`, `patches/`.
Pitfalls: nested overlays (flatten!); not pinning image tags (use `images:` with explicit `newTag`); drift between cluster + kustomization (no release state tracked, unlike Helm); skipping `kustomize build` validation before apply; using traditional ConfigMaps when rollout-on-change needed (use generator).

7.5 Helm vs. Kustomize

Helm: templating + package manager (variables via `values.yaml`, releases, repos, rollbacks); good for distributing reusable apps.
Kustomize: template-free overlays (patch/merge plain YAML per environment), built into `kubectl` (`-k`); good for own manifests across dev/test/prod. The two are often combined (Helm to package, Kustomize to patch).

8 K8s Monitoring & Logging

Observability is a property (degree system generates actionable insights, qualitative);
Observing/Monitoring is a process (collect/store/evaluate *signals*).
Three pillars:
Metrics (quantitative: util, counters, rates),
Logs (qualitative: stack traces, events),
Traces (request path across services via *spans* – latency/bottlenecks, e.g. Tempo, Jaegerfollow one request across services via *spans* to find latency/bottlenecks, e.g. Tempo, Jaeger).
Monitoring vs Logging complement each other.
K8s obs gaps: no historical metrics, `kubectl logs` only running containers (lost when container dies, kubelet cleanup), no native dashboards/alerts. Container logs at `/var/log/pods/...`; kubelet rotates at 10MB, keeps 5.
Push (agent-based): agent on system sends to collector; observed decides when.
Pull (agentless): collector queries via SNMP/ICMP/HTTP; collector decides when, but system must be online.
Golden Signals (Google SRE): latency, traffic, errors, saturation.

> **PLG stack** (lightweight, K8s-native): Prometheus+Loki+Grafana. **LGTM:** Loki, Grafana, Tempo, Mimir. **EFK/ELK:** Elasticsearch (Java, JSON index, full-text, heavy) + Kibana + Fluentd/Logstash. PLG indexes *labels* only (cheap, fast, weaker text search).
> **OpenTelemetry:** unified telemetry (metrics/logs/traces) spec.

8.1 Prometheus & PromQL

Go, TSDB, pull-based scrape over HTTP (`/metrics`). Pushgateway/remote-write for short jobs. Client libs Go/Java/Python/Ruby/Rust. Standalone (no remote dep).
Targets: Kubelets, kube-state-metrics, exporters.
Exporters: bridge for apps without native `/metrics` (PostgreSQL, NGINX, Redis, JIRA, Kafka, …); use when can't instrument.
node-exporter: HW/OS.
kube-state-metrics: K8s object state. Install `metrics-server` for `kubectl top`. CPU units: `100m`=0.1 CPU absolute.
Metric types: *Counter* (monotonic, reset to 0: requests/errors), *Gauge* (current value: memory, concurrency), *Histogram* (sampled buckets), *Summary* (sampled + total/sum).
Format: `name{label=val,…} value [ts]`, foundation of **OpenMetrics** std.
PromQL data types: Instant vector (set of series, one sample each, same ts), Range vector (range per series, e.g. `[5m]`), Scalar (float), String (unused).
Offset: (positive = backwards): `rate(x[5m] offset 1w)`.
@-modifier: `sum(x 1609746000). histogram_quantile(0.95, rate(…_bucket[5m]))` for p95 from Histogram. `rate()` useless on spikes/outliers – use raw & > threshold.

```
sum by (namespace) (kube_pod_status_ready{
      condition="false"})
rate(mysql_global_status_commands_total{command
      =~"select"}[5m])*100 > 35
```

Prometheus Operator (installed via the `kube-prometheus-stack` Helm chart): deploys operator + Prometheus + Grafana + Alertmanager in one go, and lets you configure monitoring *declaratively* through CRDs instead of editing `prometheus.yml`.
CRDs: `Prometheus` (a Prometheus instance), `ServiceMonitor` / `PodMonitor` (declare which Services/Pods to scrape).
How a target gets scraped – a chain of label selectors, each picking the next level down: `Prometheus.serviceMonitorSelector` → **ServiceMonitors**, each ServiceMonitor's `selector` → **Services**, the Service's own `selector` → **Pods**. Drop in a ServiceMonitor and its pods are auto-scraped – no Prometheus config edit.
Scaling Prometheus: *Federation* (hierarchical: region scrapes DC); *Thanos* (sidecar, global query, long-term, dedup); *Grafana Mimir* (horiz. scalable, ingester/querier components, multi-tenant, remote-write only).

8.2 Grafana, Alerting

General-purpose viz, many data sources (Prom/Loki/ES). *Explore* feature: browse data sources, build queries, debug, share. OAuth/-SAML/LDAP. Dashboard JSON repo `grafana.com/grafana/dashboards`. Tips: important data top, status-light > number, simplicity, Data Source variable.
Alertmanager vs Grafana Alerts: Alertmanager (Prom ecosystem, GitOps-friendly, separates viz/alert, Prom-only). Grafana Alerts (any data source, tied to dashboards, can chain to ext Alertmanager, less GitOps-friendly). Flow: *Rule*→ evaluate/threshold → *Alert* →Silence?→AlertManager→Channels (Email/Slack/Teams/Discord/Threema/Webhook)→Contact Points.
Warning ≠ Alert: cert expires in 45d, CPU 75%, budget 200% are warnings not pages. Avoid alert fatigue, build up slow, test.

8.3 Logging: Loki, Alloy

Architectures: *Node-level* (DaemonSet agent reads node log file – survives container death, batched); *App-level* (app pushes to backend directly); *Sidecar* (streaming = sidecar writes to file then node agent ships – `kubectl logs` won't show sidecar's own; logging-agent sidecar = ships to backend directly, low-latency but loses on disconnect, adds complexity, misses crashes). Best practice: structured JSON logs.
Loki: horiz. scalable, indexes *labels* only (not text), *nanosecond* unix ts, cheap storage, PromQL-like **LogQL**: `{job=ttest"} |= ërror";` metric queries `rate({app=*"} |= "176"[1m])`; filters `|=` contains, `!=`, `|~` regex, `!~`.
Alloy (Grafana, replaces Promtail): collects all telemetry, own HCL-like lang. Components: `loki.source.file` (tail files, classic DaemonSet pattern), `loki.source.kubernetes` (subscribe via K8s API – no root, no DaemonSet needed, but more CPU/net), `loki.source.kubernetes_events` (K8s events). Logs enriched with **K8s metadata**, forwarded over HTTP. Helm-installable.
Pitfalls: too much logging → storage/complexity; no alert tuning → noise/missed; forgetting rotation/archival; relying on single signal; logging secrets.

9 Service Mesh

9.1 Microservices

Small independent services, one business function, communicate via lightweight APIs. Microservices == distributed computing.
Pros: independent deploy, flexibility, scalability, fault isolation, team autonomy.
Cons: operational complexity, distributed-system challenges, data consistency, higher resource use.
Challenges: Communication: services must know endpoints, new service ripples changes. Security: in-cluster traffic unsecured, any pod → any pod. Monitoring: spread out, logic must be added per app.

9.2 Distributed Computing Fallacies

Network reliable; latency zero; bandwidth infinite; topology stable; one admin; transport cost zero; network homogeneous. All false → motivate mesh.

9.3 Service Mesh

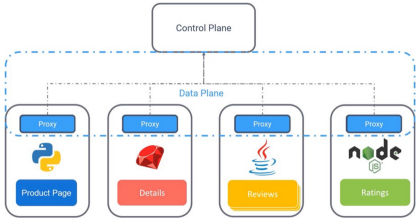
Dedicated infra layer handling service-to-service communication: traffic routing, security, observability, resiliency - abstracted from app code.
Capabilities: Traffic Mgmt (routing, splitting), Resilience (fault tolerance, retries, fallback), Security (mTLS, authN/Z), Observability (metrics, logs, tracing).
Why: centralized policy, uniform observability, faster incident resolution, transparent security, advanced rollouts (canary/AB), resilience.
Components: Data Plane: lightweight proxies (Envoy), handle comm/metrics/security. Control Plane: manages config/policies (Istiod), pushes rules to data plane.

9.4 Mesh Evolution: Library → Sidecar → Kernel

Library: mesh logic linked into each app, per-language lib (Python, Go, ...). Invasive, language-locked.
Sidecar: proxy per pod, transparent traffic interception, no code change, language-agnostic. Downsides: invasive pod-spec mutation, upgrade = pod restart, large CPU/RAM per pod (worst-case provisioning), extra hop.
Kernel/Ambient: mesh in kernel (eBPF) or per-node agent. 30 pods/-node → 1 proxy/node instead of 30. L7 still typically needs userspace proxy.
Leading meshes: Istio (Google, feature-rich), Linkerd (lightweight, perf), Kuma (CNCF, VMs+containers), Consul Connect (HashiCorp, discovery), Cilium (eBPF, not dedicated mesh).

9.5 Sidecar Pattern Pros/Cons

Pros: + transparent (no code change, lang-agnostic), + per-pod isolation, + platform team owns cross-cutting (mTLS/retries/routing/telemetry).
Cons: - CPU/RAM scales with pod count, - extra hop = latency, - sidecar injection + proxy upgrade needs restart, - overkill for few services.



Envoy proxies authenticate via **mTLS** (mutual TLS, both sides present certs).

9.6 Ambient Mesh (Sidecarless Istio)

Sidecar downsides → ambient: per-node **zTunnel** agent (L4, mTLS, authN/Z, no HTTP parsing) communicates across nodes via HTTPS CONNECT tunnel. Optional L7 **Waypoint Proxy** per

namespace (Envoy) for rich policies/telemetry. zTunnel alone = L4 metrics only.
Slicing: Secure Overlay (L4): TCP routing, mTLS tunneling, simple authZ, TCP metrics. L7 Processing: HTTP routing/LB, circuit breaking, rate limiting, fault injection, retry, timeouts, rich authZ, HTTP metrics/access logs/tracing.
Cilium variant: dynamic **eBPF** programs in kernel handle L3/L4 (network path); per-node Envoy proxy for L7 termination when needed. No sidecar, no extra hop for L4.

9.7 Gateway API & Ambient Routing / GAMMA

Enable ambient: `kubectl label ns x istio.io/dataplane-mode=ambient` (sidecar mode: `istio-injection=enabled`). Routes via K8s **Gateway API** (`HTTPRoute`) not VirtualService. Two Deployments behind one Service ≈50/50 default; override via weights. Best practice: `version` label per Deployment.
GAMMA (Gateway API for Mesh Management & Administration): Gateway API for east/west (mesh) traffic, not just north/south (ingress). Adopted by more meshes.

```
# weighted canary 90/10:
- backendRefs:
  - { name: frontend, port: 5000, weight: 90 }
  - { name: frontend-v2, port: 5000, weight: 10
      }
# mirror/shadow (copy to v2, drop response):
- backendRefs:
  - { name: frontend, port: 5000 }
  filters:
  - type: RequestMirror
    requestMirror: { backendRef: { name:
        frontend-v2, port: 5000 } }
```

istio-system pods: Istiod, Ingress Gateway (Envoy), Prometheus, Grafana, Jaeger, Kiali. **Kiali shapes:** rectangle=app, triangle=Service, square=workload/pod.

9.8 Traffic Management (Istio Classic)

Gateway replaces K8s ingress/egress. VirtualService binds gateway, routes to services. DestinationRule applies post-routing policy: LB, connection pool, outlier detection (circuit breaking), TLS mode, defines subsets (versions).

```
apiVersion: networking.istio.io/v1beta1
kind: Gateway
metadata: { name: my-gateway }
spec:
  selector: { istio: ingressgateway }
  servers:
  - port: { number: 80, name: http, protocol:
        HTTP }
    hosts: ["example.com"]
--
kind: VirtualService
metadata: { name: my-vs }
spec:
  hosts: ["example.ch"] # must match gateway
      host
  gateways: [my-gateway]
  http:
  - route:
    - destination:
        host: my-service
        weight: 99 # split traffic if 2nd dest
            defined
        port: { number: 80 }
--
kind: DestinationRule
metadata: { name: my-dr }
spec:
  host: my-service
  trafficPolicy:
    tls: { mode: ISTIO_MUTUAL }
```

9.9 Resilience & Use Cases

Fault Injection: simulate net failures, test error handling. **Timeouts:** cap request time, avoid queue buildup. **Retries:** auto-retry on failure.
Cascading Failure: one service dies → callers pile up → whole graph red.
Circuit Breaking: trip on errors/outlier → prevent cascading failure (DestinationRule outlierDetection).
Rate Limiting: cap RPS, protect downstream from overload.
Zero-Trust: encrypt + authenticate every call, no implicit trust in cluster. Istio exports Prometheus metrics by default.

9.10 Deployment Strategies

Rolling: gradual pod replace, default K8s. **Recreate / Big-bang:** kill all, deploy new (downtime). **Canary:** small % to v2, watch, ramp up. **Blue/Green:** two envs, switch traffic atomically, easy rollback. **Dark Launching / Shadow:** mirror prod traffic to v2, drop response (test under real load). See General Definitions.

9.11 Security

AuthN: mTLS between services. **AuthZ:** AuthorizationPolicy = fine-grained RBAC.

```
# enforce mTLS in namespace
kind: PeerAuthentication
metadata: { name: default, namespace:
      production }
spec:
  mtls: { mode: STRICT }
--
kind: AuthorizationPolicy
metadata: { name: require-jwt, namespace: foo }
spec:
  selector: { matchLabels: { app: httpbin } }
  action: ALLOW
  rules:
  - from: [{ source: { namespaces: ["frontend"]
        } }]
    to: [{ operation: { methods: ["GET"], paths:
          ["/headers"] } }]
```

10 Cilium & eBPF

10.1 eBPF

eBPF (extended Berkeley Packet Filter): runs sandboxed programs directly in the Linux kernel without changing kernel source or loading modules. Programs attach to hooks (syscalls, network events) and are verified before loading. Enables high-performance networking, observability and security at kernel level.

10.2 Cilium

A CNI (Container Network Interface) plugin built on eBPF. Replaces `kube-proxy`: instead of one iptables rule chain per Service, it programs eBPF maps in the kernel (faster, scales better).
Identity-based security: Cilium derives a security *identity* from a pod's labels (not from its IP). Policy is enforced on identities, so it survives pod rescheduling and IP changes.
Hubble: Cilium's observability layer; gives network flow visibility (service map, L3–L7 flows).

10.3 Network Policies

Declarative firewall rules for pod traffic. Plain `NetworkPolicy` works at L3/L4 (IP, port, protocol). `CiliumNetworkPolicy` adds L7 (HTTP method/path, DNS, gRPC, Kafka).
Default deny: as soon as a pod is selected by any policy, all non-matching traffic is dropped (whitelist model). Rules split into **ingress** (incoming) and **egress** (outgoing) and are *stateful*: return traffic of an allowed connection is permitted automatically.

```
apiVersion: cilium.io/v2
kind: CiliumNetworkPolicy
metadata:
  name: allow-frontend
spec:
  endpointSelector:
    matchLabels:
      app: backend # applies to backend pods
  ingress:
  - fromEndpoints:
    - matchLabels:
        app: frontend # only frontend may connect
    toPorts:
    - ports:
      - port: "80"
        protocol: TCP
      rules:
        http: # L7 rule
        - method: GET
          path: "/api/.*"
```

11 GitOps

Git = single source of truth for infra/app config; in-cluster agent reconciles live → desired state.
4 Principles (OpenGitOps): declarative state; versioned & immutable (history = audit, revert = rollback); pulled by agents; continuously reconciled (detect + self-heal drift).

11.1 Push vs. Pull

Push (classic CI/CD): pipeline holds creds, runs `kubectl apply`. + fast, well-known; - creds outside cluster, inbound firewall, no drift detection, per-env pipeline edits.
Pull (GitOps): in-cluster agent watches Git, syncs. + no inbound access, drift correction, multi-cluster scale; - agent per cluster, polling latency. Trust: CI server → agent + Git. Access dir: push = in; pull = out.

11.2 Argo CD vs. Flux CD

Argo CD: Web UI + CLI, multi-cluster from one UI, SSO/RBAC, sync waves, pre/post hooks, health checks. CRDs: `Application` (repo+path → cluster/ns + syncPolicy), `ApplicationSet` (templated per branch/cluster/env), `AppProject` (restrict what/where/who). Pattern: App-of-Apps (root App syncs children).
Flux CD: CLI-first, no UI, modular GitOps Toolkit (source, kustomize, helm, notification, image). Supports Gitless GitOps (OCI/S3) + image automation. Naming clash: `kustomize.config.k8s.io` (feature) vs. `kustomize.toolkit.fluxcd.io` (Flux CRD).

11.3 Secrets & Best Practices

Encrypted in Git: Sealed Secrets (controller decrypts), SOPS. **Referenced:** External Secrets, Secrets Store CSI driver (mounts Vault/KMS as volume).
Monorepo: `clusters/<c>/` bootstrap, `apps/base/` shared, `apps/<c>/` per-cluster. **Rules:** never commit plaintext, pin image tags (no `latest`), monitor controller, split monolith repos.

12 Site Reliability Engineering

"Keep the site up, whatever it takes"
SL Indicators: A measurement (e.g. Response latency over a period of 10 min) **SL Objectives:** A Objective (e.g. 99.9% below 5ms) **SL Agreements:** A economic incentive (e.g. or less bonus) **Error Budget:** 1 - SLO (e.g. 100% - 99.9% = 0.1%)

The four golden signals: latency, traffic, errors, and saturation.

12.1 Outage Handling

Goals: 1. Minimize impact (Automated, fast detection; good diagnostic information, practice remidation) 2. Prevent reoccurrence (handle event, write postmortem)